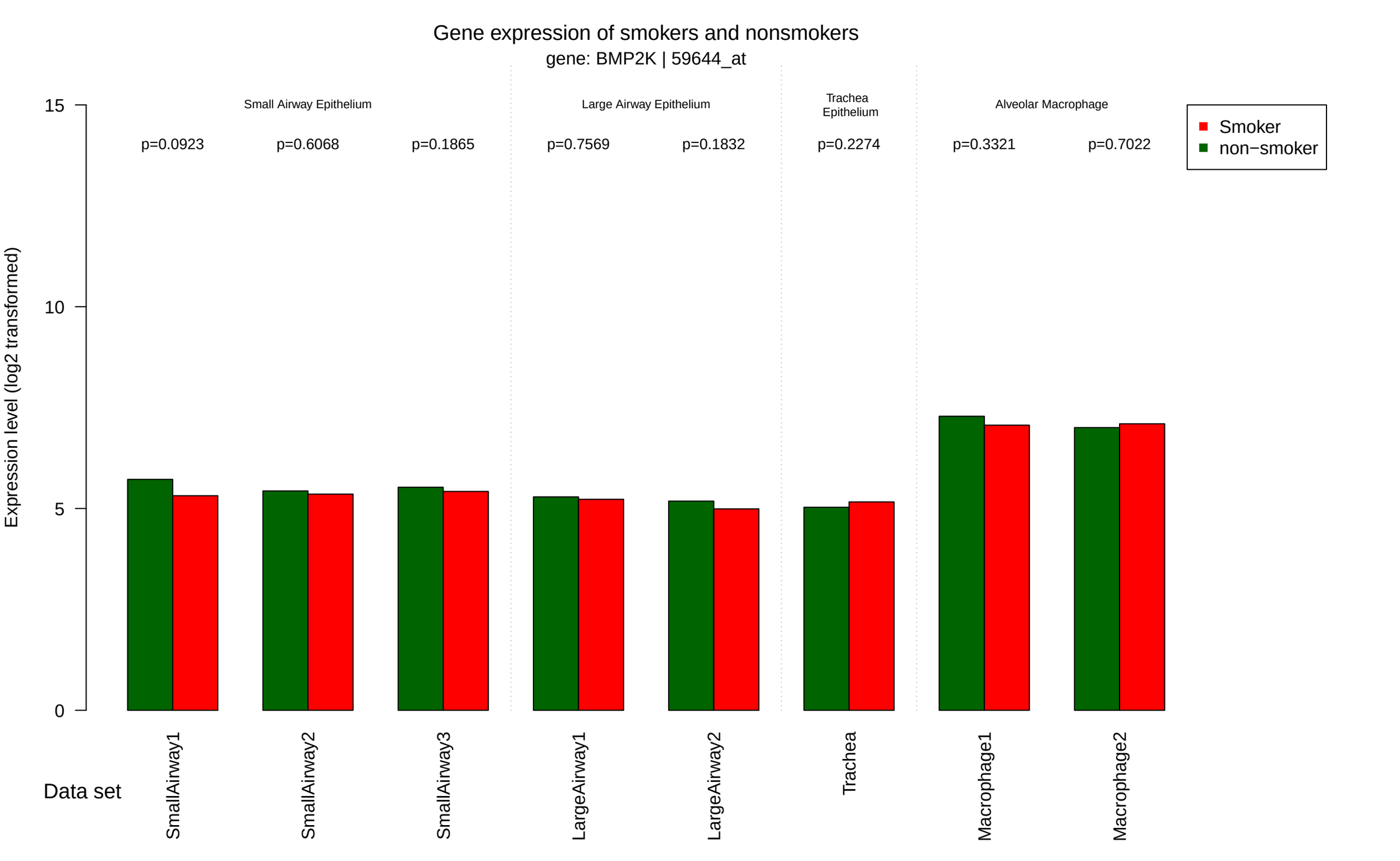
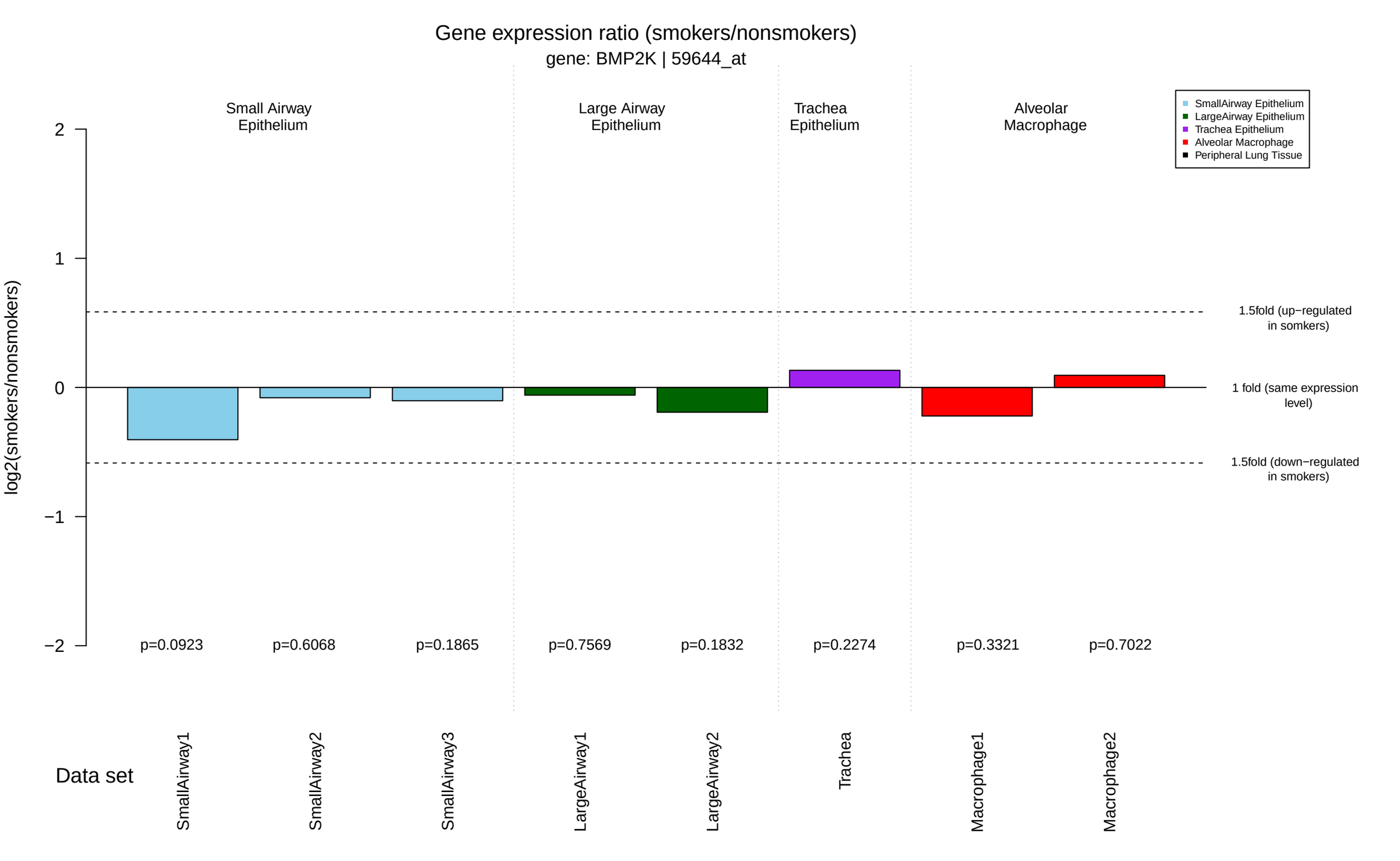
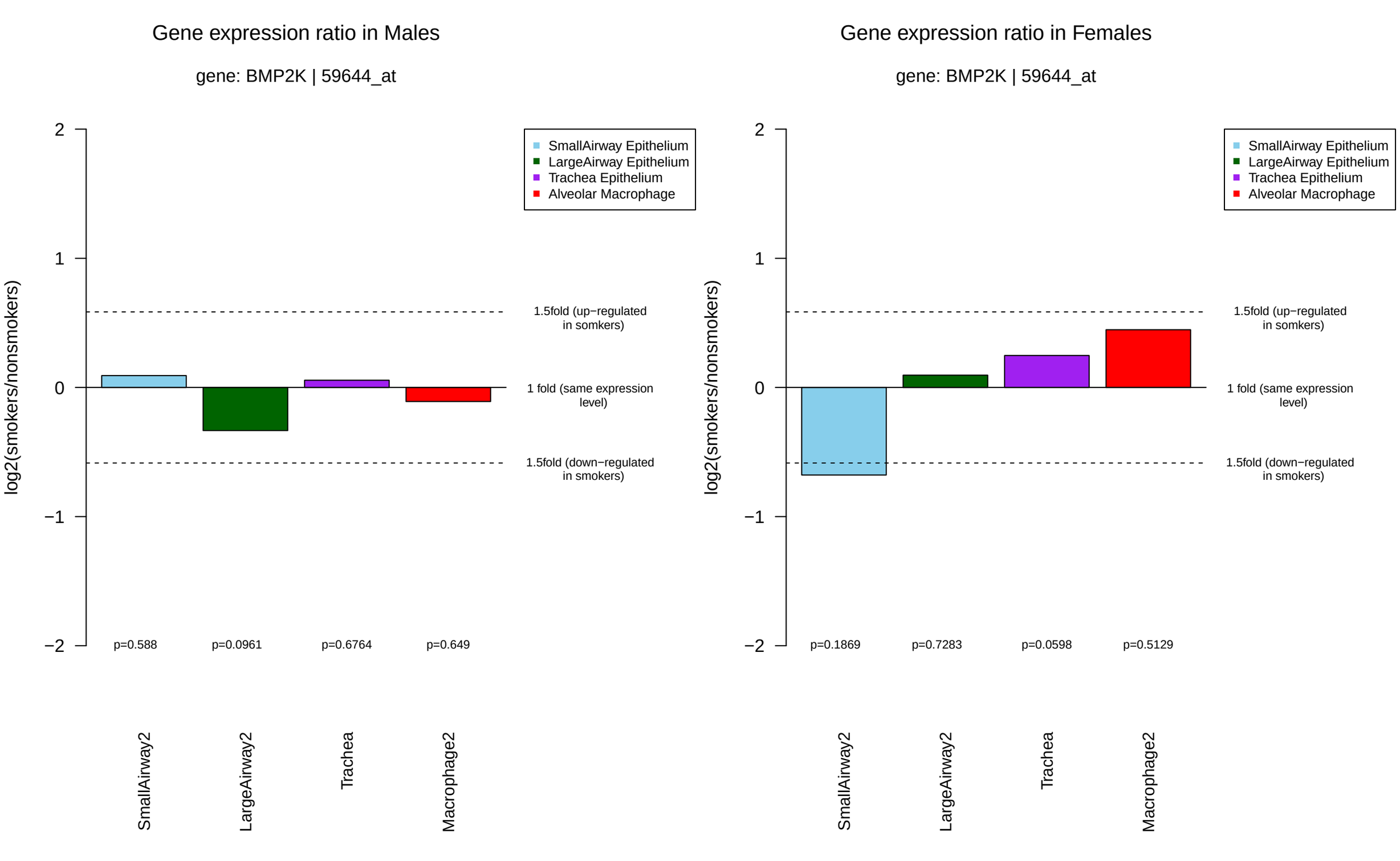
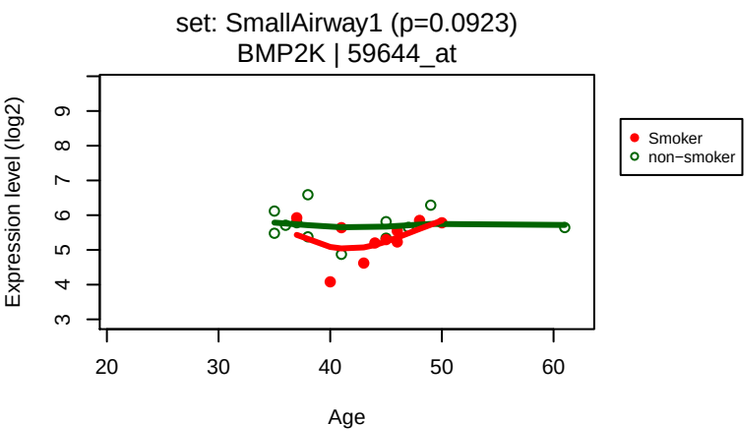
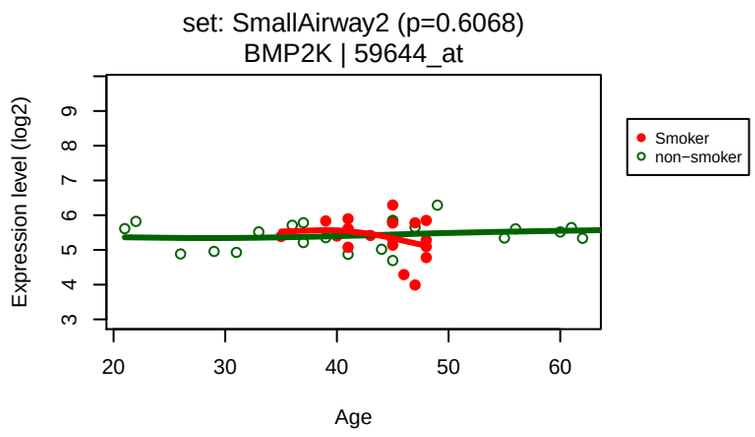
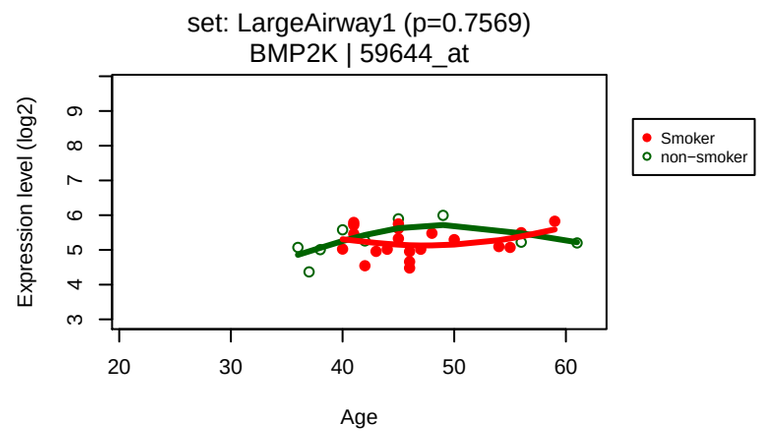

Gene expression of smokers and nonsmokers
gene: BMP2K | 59644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0923
p=0.6068
p=0.1865
p=0.7569
p=0.1832
p=0.2274
p=0.3321
p=0.7022
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

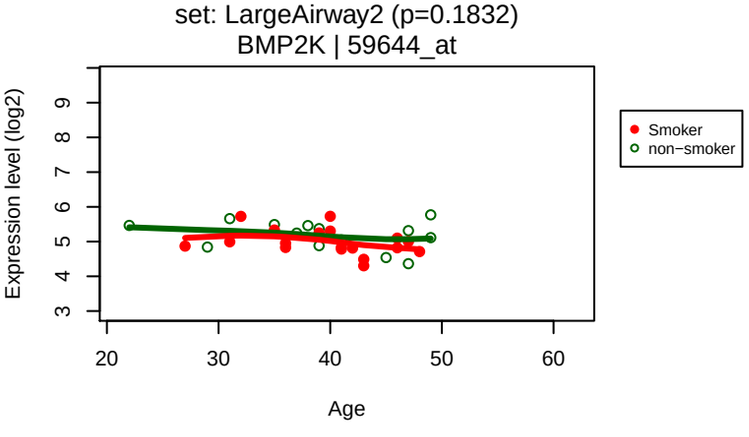

Gene expression ratio (smokers/nonsmokers)
gene: BMP2K | 59644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0923
p=0.6068
p=0.1865
p=0.7569
p=0.1832
p=0.2274
p=0.3321
p=0.7022
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

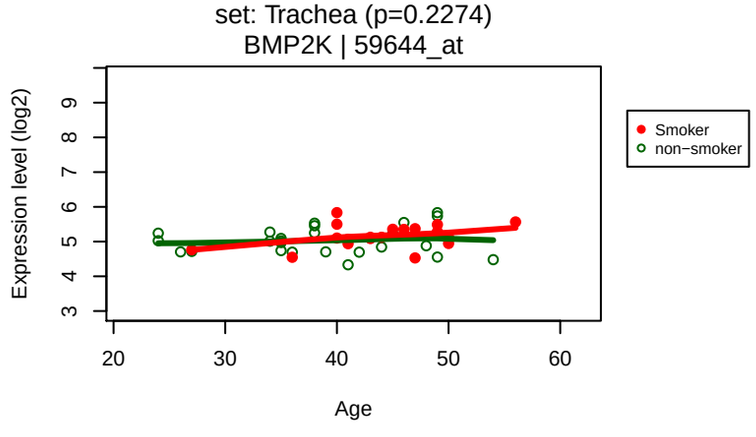

Gene expression ratio in Males
gene: BMP2K | 59644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.588
p=0.0961
p=0.6764
p=0.649
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2K | 59644_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1869
p=0.7283
p=0.0598
p=0.5129
SmallAirway2
LargeAirway2
Trachea
Macrophage2

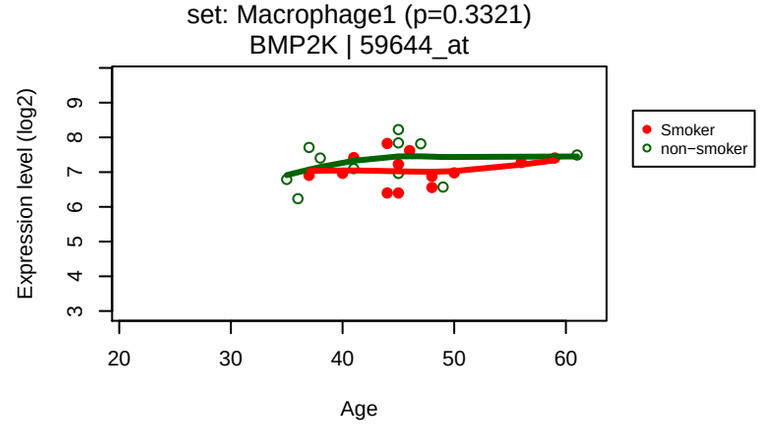

set: SmallAirway1 (p=0.0923)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker

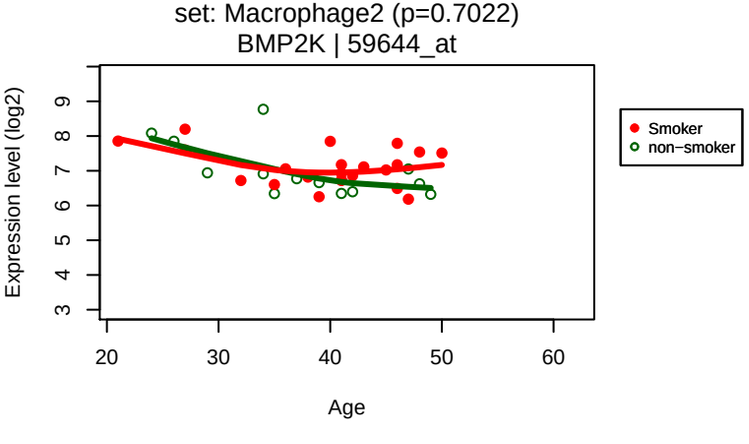

set: SmallAirway2 (p=0.6068)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker

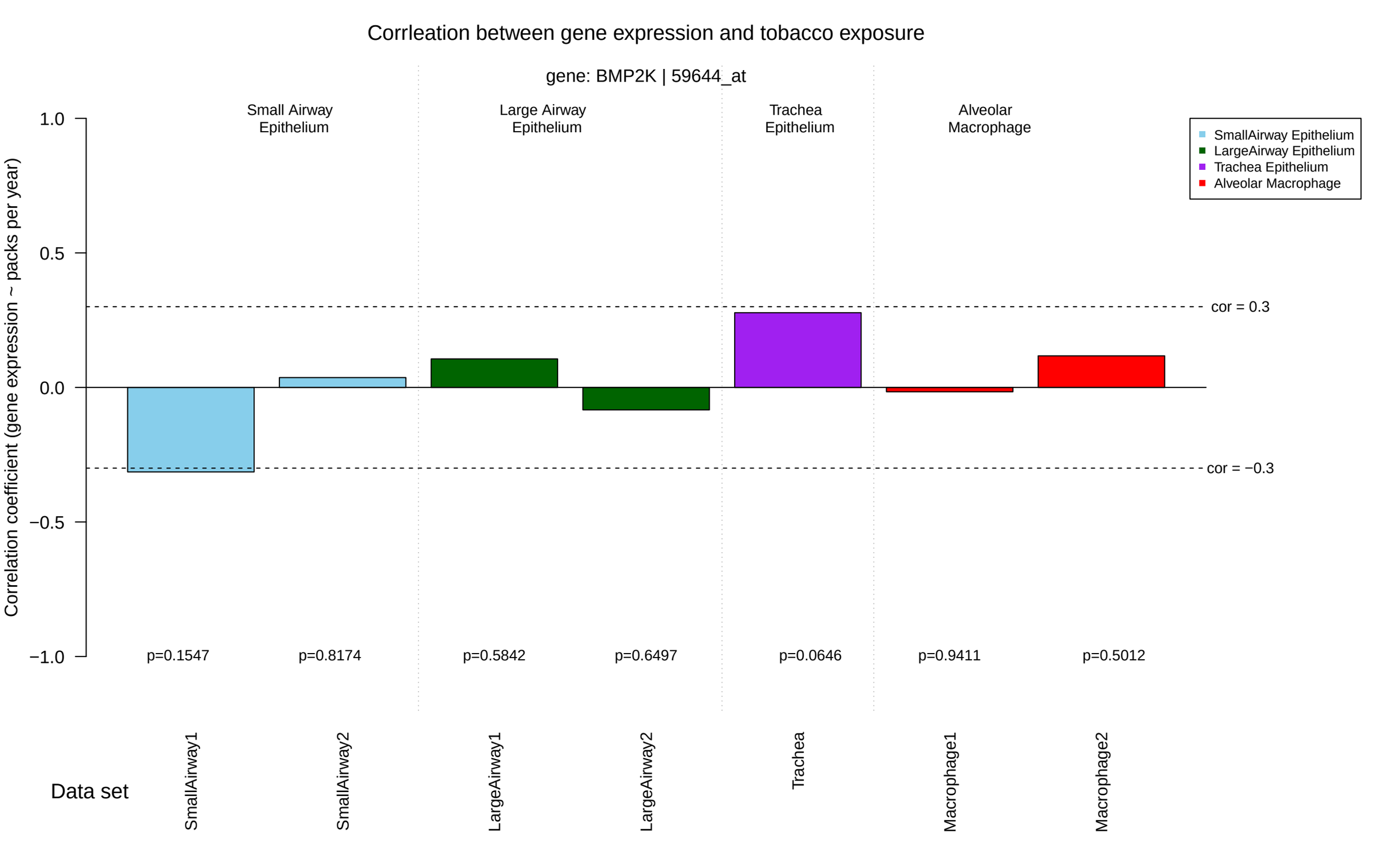

set: LargeAirway1 (p=0.7569)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1832)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2274)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3321)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7022)
BMP2K | 59644_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2K | 59644_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1547
p=0.8174
p=0.5842
p=0.6497
p=0.0646
p=0.9411
p=0.5012
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2